

Objectives

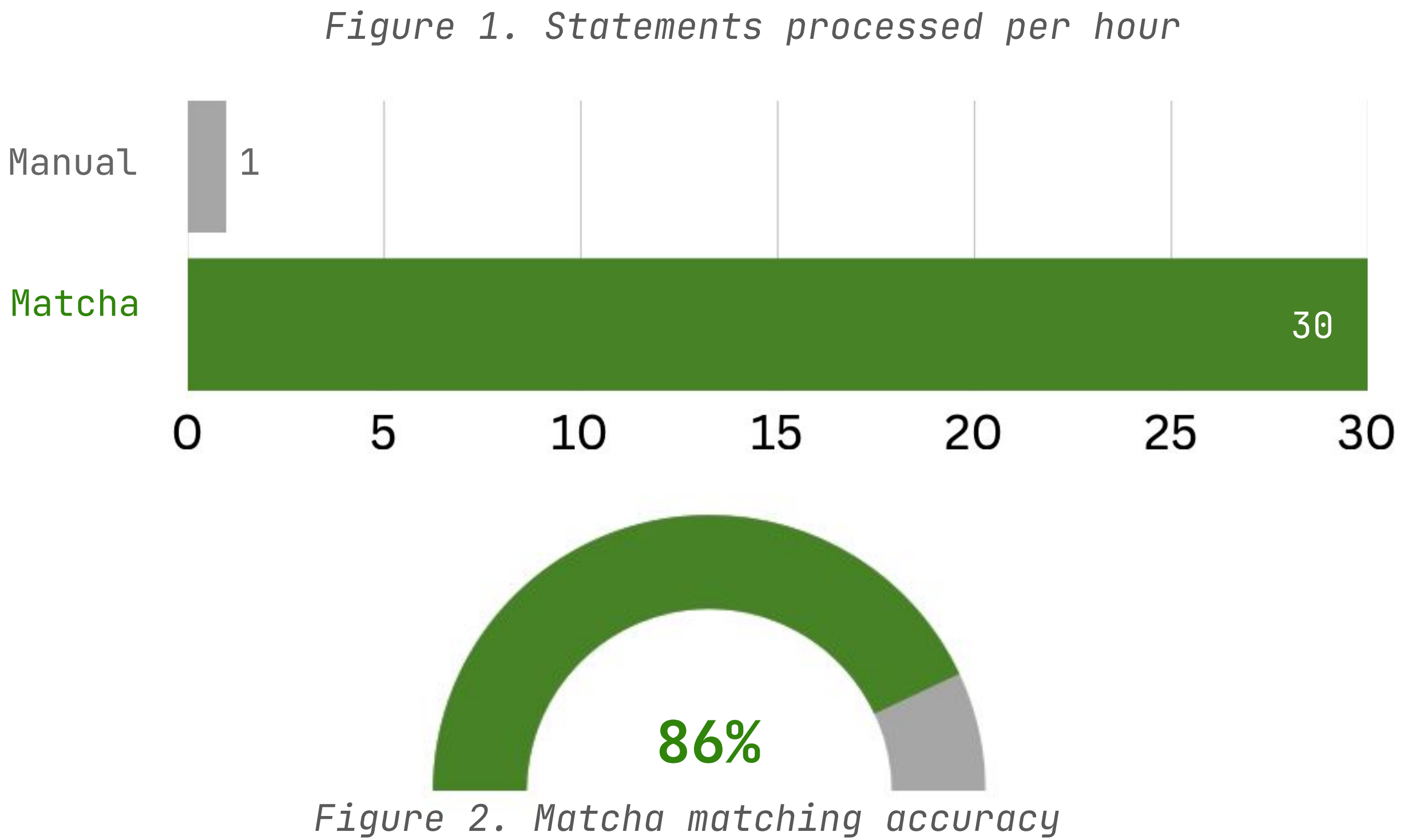
The manual bank reconciliation process is a very *time-consuming* and *error-prone* one. **Matcha** aims to automate this process with a powerful parsing system and robust matching algorithm.

Solution

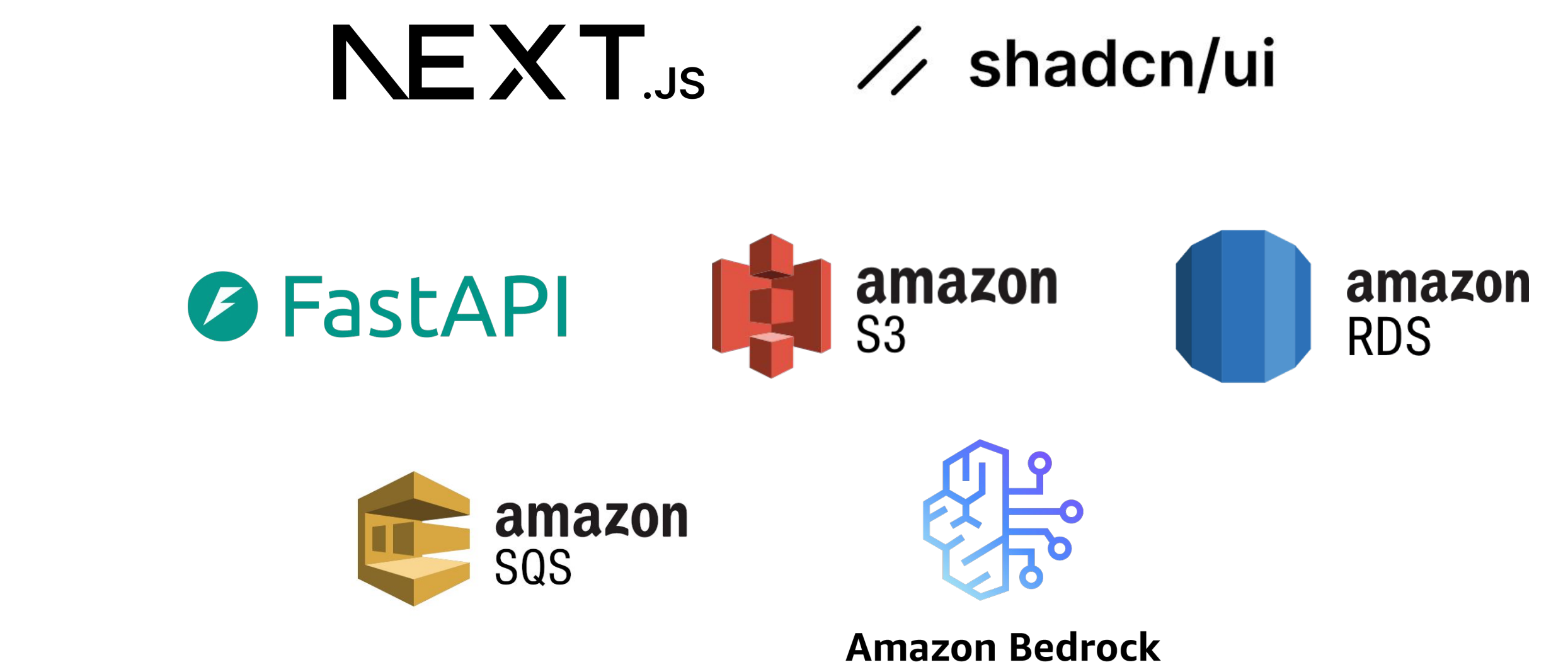
Matcha is a full-stack web application that automates the reconciliation of bank statements against supporting receipt documents.

- Uploaded statements and receipts are processed through an asynchronous cloud pipeline that uses Vision-Language Models (VLM) to extract structured data.
- A three-pass matching algorithm then reconciles statement lines to receipts via exact, fuzzy, and bundle-aggregation strategies.
- Results are served to an interactive dashboard where users can review match statuses, correct extracted fields, and manually resolve discrepancies.

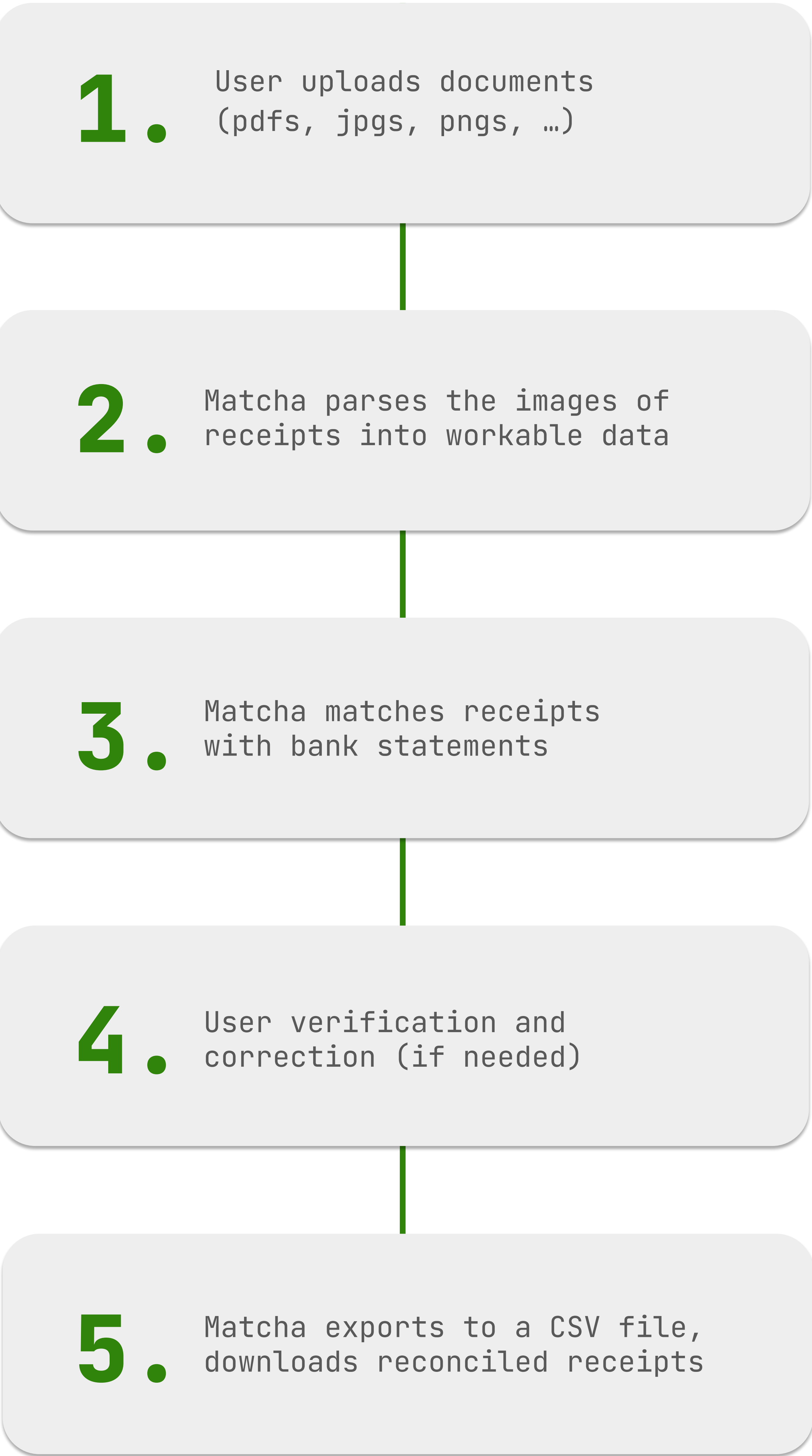
Impacts



Technology



Workflow



Future works

- **Sentence Transformer:** Improve vendor similarity score with text embedding model.
- **Universal statement parser:** Finetune in-house VLM for a more accurate and universal statement parser.
- **Real-time collaboration support:** Allow for multiple users to add comments and approve reconciliation in the same session via websocket.
- **ML-augmented matching function:** Train a reconciliation model with self updating context of matching pattern history.

Architecture

